
PRIOR AUTHORIZATION TRIAGE & POLICY COMPANION

Output Explainability Demo (mock mode)

LLM Mode : LIVE - Qwen via LM Studio

APPROVE - Covered dx + J5 state (TX) [LCD Path]

REQUEST

Procedure Code : 64483

Diagnosis : M54.16

State : TX

Patient Age : 55

Clinical Notes : Patient presents with lumbar radiculopathy confirmed on MRI. Conservative therapy was tried for 8 weeks without relief.

POLICY IDENTIFICATION

NCD : NOT_ADDRESSED

Jurisdiction : TX -> MATCHED

LCD : L39054 Epidural Injections for Pain Management

Article : A12345

POLICY EVIDENCE

Source : LCD L39054

Section : Indications

Text : Epidural steroid injections (ESIs) are covered when the patient has a diagnosis of radiculopathy (ICD-10: M54.16, M54.17) confirmed by clinical examination and supported by imaging (MRI or CT). The pr

Similarity : 0.9000

Source : LCD L39054

Section : Limitations

Text : The following diagnoses are non-covered under this LCD: routine general examination (Z00.00), encounters without clinical indication of spinal pathology. Injections performed solely for general pain w

Similarity : 0.9000

Source : LCD L39054

Section : Documentation Requirements

Text : Documentation must support the reasonable and necessary nature of the epidural injection. Required elements: (1) radiculopathy symptoms confirmed on MRI or CT imaging, (2) failure of conservative ther

Similarity : 0.9000

CODE MATCHING

HCPCS 64483 -> Article A12345 -> MATCHED

ICD-10 M54.16 -> Article A12345 -> COVERED

POLICY CRITERIA

C1 [SQL] LCD-L39054-C7f7eb3

Requirement : patient has a diagnosis of radiculopathy (ICD-10: M54

Type : STRUCTURED

Result : SATISFIED

Policy Ev. : patient has a diagnosis of radiculopathy (ICD-10: M54; LCD L39054 -> ICD-10 covered: M54.16

Patient Ev. : Submitted ICD-10: M54.16

Explanation : The submitted diagnosis code(s) M54.16 are present in the LCD L39054 covered ICD-10 data. The diagnosis requirement is satisfied.

C2 [AGENTIC_QWEN] LCD-L39054-C8ff2d4

Requirement : The following diagnoses are non-covered under this LCD: routine general examination (Z00.00), encounters witho...

Type : SEMANTIC

Result : UNKNOWN

Policy Ev. : Required: confirmed_structural_diagnosis: Evidence that the patient has a confirmed structural diagnosis of spinal pathology, such as a specific diagnosis code (e.g., M54.16) indicating a structural condition; Required:

clinical_indication_of_spinal_pathology: Documentation showing clinical indication of spinal pathology, such as physical exam findings, imaging results, or other clinical evidence supporting a specific spinal condition

Patient Ev. : Patient presents with lumbar radiculopathy confirmed on Magnetic Resonance Imaging (MRI).

Explanation : =====

AGENTIC SEMANTIC EVALUATION

Criterion: The following diagnoses are non-covered under this LCD: routine general examination (Z00.00), encounters without clinical indication of spinal pathology. Injections performed solely for general pain without a confirmed structural diagnosis are not covered. Maximum of 3 injections per 6-month period unless clinical documentation supports additional treatment.

Policy: LCD L39054

Required Evidence:

* confirmed_structural_diagnosis: Evidence that the patient has a confirmed structural diagnosis of spinal pathology, such as a specific diagnosis code (e.g., M54.16) indicating a structural condition

* clinical_indication_of_spinal_pathology: Documentation showing clinical indication of spinal pathology, such as physical exam findings, imaging results, or other clinical evidence supporting a specific spinal condition

* clinical_documentation_supporting_additional_injections: Clinical documentation that supports additional injections beyond three per six months, such as a detailed treatment plan or medical justification

Patient Evidence:

* Patient presents with lumbar radiculopathy confirmed on Magnetic Resonance Imaging (MRI).

Missing Evidence:

* No confirmed structural diagnosis is explicitly stated with a specific diagnosis code (e.g., M54.16) in the context of a structural spinal pathology.

* No clinical documentation supporting additional injections beyond three per six months is provided.

* The diagnosis M54.16 is listed in diagnoses but not explicitly linked to a confirmed structural diagnosis of spinal pathology in the clinical context.

Qwen Result: UNKNOWN

Critic Result: VALIDATED

Final Result: UNKNOWN

Insufficient or inconclusive evidence to determine whether this semantic criterion is satisfied. The result is UNKNOWN. Deterministic rules and EvidenceFusion will determine final coverage.

(Agentic pipeline completed in 64072ms)

Agent Trace:

[POLICY_AGENT] COMPLETED: Identified 3 required evidence categories in 26153ms.
[CLINICAL_EVIDENCE_AGENT] COMPLETED: Found 1 supporting, 0 contradicting, 3 missing evidence items in 23487ms.
[EVALUATION_AGENT] COMPLETED: Partial evidence: 1 supporting items found, but 3 required evidence categories are missing.
Evidence is insufficient for a definitive conclusion.
[QWEN] COMPLETED: Qwen returned UNKNOWN. Evidence cited: 1 items.
[CRITIC_AGENT] COMPLETED: Critic VALIDATED Qwen's result: UNKNOWN. All 6 checks passed.

C3 [AGENTIC_QWEN] LCD-L39054-C7eca10

Requirement : Documentation must support the reasonable and necessary nature of the epidural injection
Type : SEMANTIC
Result : UNKNOWN
Policy Ev. : Required: diagnosis: Documentation of the patient's diagnosis, such as M54.16, that supports the indication for epidural injection
Patient Ev. : Patient presents with lumbar radiculopathy confirmed on Magnetic Resonance Imaging (MRI).; Conservative therapy was tried for 8 weeks without relief.
Explanation : =====
AGENTIC SEMANTIC EVALUATION

Criterion: Documentation must support the reasonable and necessary nature of the epidural injection
Policy: LCD L39054

Required Evidence:
* diagnosis: Documentation of the patient's diagnosis, such as M54.16, that supports the indication for epidural injection

Patient Evidence:
* Patient presents with lumbar radiculopathy confirmed on Magnetic Resonance Imaging (MRI).
* Conservative therapy was tried for 8 weeks without relief.

Missing Evidence:
* Documentation of the patient's diagnosis, such as M54.16, that supports the indication for epidural injection

Qwen Result: UNKNOWN
Critic Result: VALIDATED
Final Result: UNKNOWN

Insufficient or inconclusive evidence to determine whether this semantic criterion is satisfied. The result is UNKNOWN. Deterministic rules and EvidenceFusion will determine final coverage.

(Agentic pipeline completed in 38923ms)

Agent Trace:
[POLICY_AGENT] COMPLETED: Identified 1 required evidence categories in 11331ms.
[CLINICAL_EVIDENCE_AGENT] COMPLETED: Found 2 supporting, 0 contradicting, 1 missing evidence items in 16150ms.
[EVALUATION_AGENT] COMPLETED: Partial evidence: 2 supporting items found, but 1 required evidence categories are missing.
Evidence is insufficient for a definitive conclusion.
[QWEN] COMPLETED: Qwen returned UNKNOWN. Evidence cited: 2 items.
[CRITIC_AGENT] COMPLETED: Critic VALIDATED Qwen's result: UNKNOWN. All 6 checks passed.

C4 [SQL] ARTICLE-A12345-HCPCS

Requirement : The requested procedure must be an applicable service under the Article.
Type : STRUCTURED
Result : SATISFIED
Policy Ev. : Article A12345 contains HCPCS 64483 in its coverage list.

Patient Ev. : Submitted HCPCS: 64483
Explanation : Evaluated as SATISFIED by SQL.

C5 [SQL] ARTICLE-A12345-ICD10-M54.16
Requirement : The patient's diagnosis must be an eligible diagnosis under the Article.
Type : STRUCTURED
Result : SATISFIED
Policy Ev. : Diagnosis M54.16 is present in the Article's covered ICD-10 data.
Patient Ev. : Submitted ICD-10: M54.16
Explanation : Evaluated as SATISFIED by SQL.

EVIDENCE FUSION

Criteria SATISFIED : 3
Criteria NOT_SATISFIED : 0
Criteria UNKNOWN : 2
Evidence Fusion Result : COVERED

FINAL DECISION

[APPROVE] Decision : APPROVE
Reason : All applicable policy criteria were satisfied. The submitted procedure and diagnosis codes are covered under the applicable Article. [ARTICLE_CRITERIA_SATISFIED]
Reason Codes: ARTICLE_CRITERIA_SATISFIED

Decision Basis:

All mandatory policy criteria were satisfied.
* LCD-L39054-C7f7eb3: SATISFIED
* LCD-L39054-C8ff2d4: UNKNOWN
* LCD-L39054-C7eca10: UNKNOWN
* ARTICLE-A12345-HCPCS: SATISFIED
* ARTICLE-A12345-ICD10-M54.16: SATISFIED
Evidence Fusion: COVERED
DecisionEngine: COVERED -> APPROVE

PEND - Explicitly non-covered diagnosis [LCD Path]

REQUEST

Procedure Code : 64483
Diagnosis : Z00.00
State : TX
Patient Age : 40
Clinical Notes : Routine general examination. No structural pathology identified.

POLICY IDENTIFICATION

NCD : NOT_ADDRESSED
Jurisdiction : TX -> MATCHED
LCD : L39054 Epidural Injections for Pain Management

POLICY EVIDENCE

Source : LCD L39054
Section : Indications
Text : Epidural steroid injections (ESIs) are covered when the patient has a diagnosis of radiculopathy (ICD-10: M54.16, M54.17) confirmed by clinical examination and supported by imaging (MRI or CT). The pr
Similarity : 0.9000
Source : LCD L39054
Section : Limitations
Text : The following diagnoses are non-covered under this LCD: routine general examination (Z00.00), encounters without clinical indication of spinal pathology. Injections performed solely for general pain w
Similarity : 0.9000
Source : LCD L39054
Section : Documentation Requirements
Text : Documentation must support the reasonable and necessary nature of the epidural injection. Required elements: (1) radiculopathy symptoms confirmed on MRI or CT imaging, (2) failure of conservative ther
Similarity : 0.9000

POLICY CRITERIA

C1 [SQL] LCD-L39054-Cb338a7
Requirement : patient has a diagnosis of radiculopathy (ICD-10: M54
Type : STRUCTURED
Result : NOT_SATISFIED
Policy Ev. : patient has a diagnosis of radiculopathy (ICD-10: M54; LCD L39054 -> ICD-10 non-covered: Z00.00
Patient Ev. : Submitted ICD-10: Z00.00
Explanation : The submitted diagnosis code(s) Z00.00 are explicitly listed in the LCD L39054 non-covered ICD-10 data. The diagnosis requirement is not satisfied.

C2 [AGENTIC_QWEN] LCD-L39054-C77f484
Requirement : The following diagnoses are non-covered under this LCD: routine general examination (Z00.00), encounters witho...
Type : SEMANTIC
Result : UNKNOWN
Policy Ev. : Required: diagnosis_with_structural_pathology: Evidence that the patient has a confirmed structural diagnosis of spinal pathology, such as degenerative disc disease, herniated disc, or spinal stenosis; Required: injection_frequency_justification: Clinical documentation supporting the need for more than three injections within a 6-month period
Explanation : =====
AGENTIC SEMANTIC EVALUATION

Criterion: The following diagnoses are non-covered under this LCD: routine general examination (Z00.00), encounters without clinical indication of spinal pathology. Injections performed solely for general pain without a confirmed structural diagnosis are not covered. Maximum of 3 injections per 6-month period unless clinical documentation supports additional treatment.
Policy: LCD L39054

Required Evidence:

- * diagnosis_with_structural_pathology: Evidence that the patient has a confirmed structural diagnosis of spinal pathology, such as degenerative disc disease, herniated disc, or spinal stenosis
- * injection_frequency_justification: Clinical documentation supporting the need for more than three injections within a 6-month period

Patient Evidence:

- * (No supporting evidence found in request)

Missing Evidence:

* No confirmed structural diagnosis of spinal pathology is stated; the patient data explicitly states 'No structural pathology identified.'
This fails to meet the requirement for [diagnosis_with_structural_pathology].
* No clinical documentation supporting more than three injections within a 6-month period is provided; therefore, [injection_frequency_justification] is missing.

Qwen Result: NOT_SATISFIED
Critic Result: REJECTED
Final Result: UNKNOWN

Insufficient or inconclusive evidence to determine whether this semantic criterion is satisfied. The result is UNKNOWN. Deterministic rules and EvidenceFusion will determine final coverage.

(Agentic pipeline completed in 49553ms)

Agent Trace:
[POLICY_AGENT] COMPLETED: Identified 2 required evidence categories in 23711ms.
[CLINICAL_EVIDENCE_AGENT] COMPLETED: Found 0 supporting, 0 contradicting, 2 missing evidence items in 17300ms.
[EVALUATION_AGENT] COMPLETED: No supporting evidence found. 2 required evidence categories are missing.
[QWEN] COMPLETED: Qwen returned NOT_SATISFIED. Evidence cited: 1 items.
[CRITIC_AGENT] COMPLETED: Critic REJECTED Qwen's result (NOT_SATISFIED). Final result: UNKNOWN. Reasons: Qwen returned NOT_SATISFIED when no contradicting evidence is present - only missing evidence. The correct result should be UNKNOWN, not NOT_SATISFIED (absence of evidence evidence of absence).

C3 [AGENTIC_QWEN] LCD-L39054-Cb0e914
Requirement : Documentation must support the reasonable and necessary nature of the epidural injection
Type : SEMANTIC
Result : UNKNOWN
Policy Ev. : Required: clinical_indication: Documentation of a diagnosed condition that warrants epidural injection, such as spinal nerve root compression or radiculopathy
Explanation : =====
AGENTIC SEMANTIC EVALUATION

Criterion: Documentation must support the reasonable and necessary nature of the epidural injection
Policy: LCD L39054

Required Evidence:
* clinical_indication: Documentation of a diagnosed condition that warrants epidural injection, such as spinal nerve root compression or radiculopathy

Patient Evidence:
* (No supporting evidence found in request)

Missing Evidence:
* Documentation of a diagnosed condition that warrants epidural injection, such as spinal nerve root compression or radiculopathy

Qwen Result: UNKNOWN
Critic Result: VALIDATED
Final Result: UNKNOWN

Insufficient or inconclusive evidence to determine whether this semantic criterion is satisfied. The result is UNKNOWN. Deterministic rules and EvidenceFusion will determine final coverage.

(Agentic pipeline completed in 25677ms)

Agent Trace:

[POLICY_AGENT] COMPLETED: Identified 1 required evidence categories in 10411ms.
[CLINICAL_EVIDENCE_AGENT] COMPLETED: Found 0 supporting, 0 contradicting, 1 missing evidence items in 9202ms.
[EVALUATION_AGENT] COMPLETED: No supporting evidence found. 1 required evidence categories are missing.
[QWEN] COMPLETED: Qwen returned UNKNOWN. Evidence cited: 0 items.
[CRITIC_AGENT] COMPLETED: Critic VALIDATED Qwen's result: UNKNOWN. All 6 checks passed.

EVIDENCE FUSION

Criteria SATISFIED : 0
Criteria NOT_SATISFIED : 1
Criteria UNKNOWN : 2
Evidence Fusion Result : EXCLUDED

FINAL DECISION

[PEND] Decision : PEND
Reason : The submitted procedure is explicitly excluded by the applicable Local Coverage Determination (LCD). The request is
pending for manual review. [LCD_EXCLUDES_PROCEDURE]
Reason Codes: LCD_EXCLUDES_PROCEDURE

Decision Basis:
The request was pended because one or more mandatory policy criteria were not satisfied or evidence was ambiguous.
* LCD-L39054-Cb338a7: NOT_SATISFIED
* LCD-L39054-C77f484: UNKNOWN
* LCD-L39054-Cb0e914: UNKNOWN
Evidence Fusion: EXCLUDED
DecisionEngine: EXCLUDED -> PEND

APPROVE - Stem Cell Transplant covered by NCD 110.23 [NCD RAG Path]

REQUEST

Procedure Code : 38240
Diagnosis : C91.10
State : TX
Patient Age : 52
Clinical Notes : Patient diagnosed with chronic lymphocytic leukemia (CLL), relapsed after first-line therapy. Allogeneic
hematopoietic stem cell transplantation recommended as curative intent option. Patient has good performance status and adequate
cardiopulmonary function documented.

POLICY IDENTIFICATION

NCD : NCD-110.23
Jurisdiction : TX -> NOT_ADDRESSED

POLICY EVIDENCE

Source : NCD NCD-110.23
Section : Coverage Indications

Text : Allogeneic hematopoietic stem cell transplantation (allo-HSCT) is covered for Medicare beneficiaries with the following conditions when reasonable and necessary: Leukemia, Aplastic Anemia, Severe Comb

Similarity : 0.9000

Source : NCD NCD-110.23

Section : Coverage with Evidence Development

Text : Allo-HSCT is covered for Medicare beneficiaries with the following indications ONLY when participating in approved prospective clinical studies meeting CED criteria: Myelodysplastic Syndromes, Multipl

Similarity : 0.9000

Source : NCD NCD-110.23

Section : Documentation Requirements

Text : Documentation to support the reasonable and necessary nature of allo-HSCTs must speak to the serious illness of the patient, reasons why the patient is considered to have relapsed or refractory diseases

Similarity : 0.9000

CODE MATCHING

HCPCS 38240 -> NCD NCD-110.23 -> MATCHED

POLICY CRITERIA

C1 [SQL] NCD-NCD-110.23-HCPCS

Requirement : The requested procedure must be an applicable service under NCD NCD-110.23.

Type : STRUCTURED

Result : SATISFIED

Policy Ev. : NCD NCD-110.23 contains HCPCS 38240 in its covered-procedure list.

Patient Ev. : Submitted HCPCS: 38240

Explanation : Procedure 38240 is listed in NCD NCD-110.23 covered HCPCS codes. Criterion SATISFIED by deterministic SQL check.

C2 [AGENTIC_QWEN] NCD-NCD-110.23-C6aea8a

Requirement : is covered for Medicare beneficiaries with the following conditions when reasonable and necessary: Leukemia, A...

Type : SEMANTIC

Result : SATISFIED

Policy Ev. : Required: diagnosis: Patient must have a diagnosis of Leukemia, Aplastic Anemia, Severe Combined Immunodeficiency disease (SCID), or Wiskott-Aldrich Syndrome

Patient Ev. : Patient diagnosed with chronic lymphocytic leukemia (Chronic Lymphocetric Leukemia (CLL)); CLL = Chronic Lymphocytic Leukemia = Leukemia

Explanation : =====

AGENTIC SEMANTIC EVALUATION

Criterion: is covered for Medicare beneficiaries with the following conditions when reasonable and necessary: Leukemia, Aplastic Anemia, Severe Combined Immunodeficiency disease (SCID), and Wiskott-Aldrich Syndrome

Policy: NCD NCD-110.23

Required Evidence:

* diagnosis: Patient must have a diagnosis of Leukemia, Aplastic Anemia, Severe Combined Immunodeficiency disease (SCID), or Wiskott-Aldrich Syndrome

Patient Evidence:

* Patient diagnosed with chronic lymphocytic leukemia (Chronic Lymphocetric Leukemia (CLL))

* CLL = Chronic Lymphocytic Leukemia = Leukemia

Qwen Result: SATISFIED

Critic Result: VALIDATED
Final Result: SATISFIED

The submitted clinical documentation satisfies this semantic policy criterion. The agentic evaluation chain VALIDATED the result.

(Agentic pipeline completed in 40708ms)

Agent Trace:
[POLICY_AGENT] COMPLETED: Identified 1 required evidence categories in 15343ms.
[CLINICAL_EVIDENCE_AGENT] COMPLETED: Found 2 supporting, 0 contradicting, 0 missing evidence items in 13503ms.
[EVALUATION_AGENT] COMPLETED: Supporting evidence found (2 items). No missing evidence categories. Policy requirement may be satisfied.
[QWEN] COMPLETED: Qwen returned SATISFIED. Evidence cited: 2 items.
[CRITIC_AGENT] COMPLETED: Critic VALIDATED Qwen's result: SATISFIED. All 6 checks passed.

C3 [AGENTIC_QWEN] NCD-NCD-110.23-Cc64785
Requirement : is covered for Medicare beneficiaries with the following indications ONLY when participating in approved prosp...
Type : SEMANTIC
Result : UNKNOWN
Policy Ev. : Required: diagnosis: Patient must have one of the following diagnoses: Myelodysplastic Syndromes, Multiple myeloma (Durie-Salmon Stage II or III), Myelofibrosis with DIPSS-plus intermediate-2 or High risk, or Sickle cell disease that is severe and symptomatic; Required: clinical_study_participation: Patient must be participating in an approved prospective clinical study that meets CED criteria
Explanation : =====
AGENTIC SEMANTIC EVALUATION

Criterion: is covered for Medicare beneficiaries with the following indications ONLY when participating in approved prospective clinical studies meeting CED criteria: Myelodysplastic Syndromes, Multiple myeloma (Durie-Salmon Stage II or III), Myelofibrosis with DIPSS-plus intermediate-2 or High risk, Sickle cell disease that is severe and symptomatic
Policy: NCD NCD-110.23

Required Evidence:
* diagnosis: Patient must have one of the following diagnoses: Myelodysplastic Syndromes, Multiple myeloma (Durie-Salmon Stage II or III), Myelofibrosis with DIPSS-plus intermediate-2 or High risk, or Sickle cell disease that is severe and symptomatic
* clinical_study_participation: Patient must be participating in an approved prospective clinical study that meets CED criteria

Patient Evidence:
* (No supporting evidence found in request)

Missing Evidence:
* Patient must have one of the following diagnoses: Myelodysplastic Syndromes, Multiple myeloma (Durie-Salmon Stage II or III), Myelofibrosis with DIPSS-plus intermediate-2 or High risk, or Sickle cell disease that is severe and symptomatic
* Patient must be participating in an approved prospective clinical study that meets CED criteria

Qwen Result: UNKNOWN
Critic Result: VALIDATED
Final Result: UNKNOWN

Insufficient or inconclusive evidence to determine whether this semantic criterion is satisfied. The result is UNKNOWN. Deterministic rules and EvidenceFusion will determine final coverage.

(Agentic pipeline completed in 61312ms)

Agent Trace:

[POLICY_AGENT] COMPLETED: Identified 2 required evidence categories in 22969ms.
[CLINICAL_EVIDENCE_AGENT] FAILED: Clinical evidence extraction failed (timed out). All required evidence reported as missing.
[EVALUATION_AGENT] COMPLETED: No supporting evidence found. 2 required evidence categories are missing.
[QWEN] COMPLETED: Qwen returned UNKNOWN. Evidence cited: 0 items.
[CRITIC_AGENT] COMPLETED: Critic VALIDATED Qwen's result: UNKNOWN. All 6 checks passed.

C4 [RULES] NCD-NCD-110.23-C8d80bc
Requirement : Documentation to support the reasonable and necessary nature of allo-HSCTs must speak to the serious illness o...
Type : RULE_BASED
Result : UNKNOWN
Policy Ev. : Documentation to support the reasonable and necessary nature of allo-HSCTs must speak to the serious illness of the patient, reasons why the patient is considered to have relapsed or refractory disease, relevant clinical contextual information such as age, frailty, performance status, cardiopulmonary function, and psychosocial support structures.
Patient Ev. : No explicit deterministic rule logic implemented for this text.
Explanation : A rule-based requirement was identified but no matching evaluation logic exists for this criterion text. The requirement cannot be evaluated.

EVIDENCE FUSION

Criteria SATISFIED : 2
Criteria NOT_SATISFIED : 0
Criteria UNKNOWN : 2
Evidence Fusion Result : UNKNOWN

FINAL DECISION

[PEND] Decision : PEND
Reason : Policy evidence was ambiguous or conflicting. Manual clinical review is required to determine coverage.
[AMBIGUOUS_EVIDENCE_REQUIRES_REVIEW]
Reason Codes: AMBIGUOUS_EVIDENCE_REQUIRES_REVIEW

Decision Basis:
The request was pended because one or more mandatory policy criteria were not satisfied or evidence was ambiguous.
* NCD-NCD-110.23-HCPCS: SATISFIED
* NCD-NCD-110.23-C6aea8a: SATISFIED
* NCD-NCD-110.23-Cc64785: UNKNOWN
* NCD-NCD-110.23-C8d80bc: UNKNOWN
Evidence Fusion: UNKNOWN
DecisionEngine: UNKNOWN -> PEND

APPROVE - AFP lab test covered by NCD 190.25 [NCD RAG Path]

REQUEST

Procedure Code : 82105
Diagnosis : C22.0
State : CA
Patient Age : 67
Clinical Notes : Hepatocellular carcinoma in high-risk patient with alcoholic cirrhosis. AFP serum test ordered to monitor response to treatment.

POLICY IDENTIFICATION

NCD : NCD-190.25
Jurisdiction : CA -> NOT_ADDRESSED

POLICY EVIDENCE

Source : NCD NCD-190.25
Section : Indications
Text : AFP (Alpha-fetoprotein) testing (HCPCS 82105, 82106) is covered for the diagnosis of hepatocellular carcinoma in high-risk patients such as those with alcoholic cirrhosis, cirrhosis of viral etiology,
Similarity : 0.9000

CODE MATCHING

HCPCS 82105 -> NCD NCD-190.25 -> MATCHED

POLICY CRITERIA

C1 [SQL] NCD-NCD-190.25-HCPCS

Requirement : The requested procedure must be an applicable service under NCD NCD-190.25.
Type : STRUCTURED
Result : SATISFIED
Policy Ev. : NCD NCD-190.25 contains HCPCS 82105 in its covered-procedure list.
Patient Ev. : Submitted HCPCS: 82105
Explanation : Procedure 82105 is listed in NCD NCD-190.25 covered HCPCS codes. Criterion SATISFIED by deterministic SQL check.

C2 [AGENTIC_QWEN] NCD-NCD-190.25-C1474c5

Requirement : is covered for the diagnosis of hepatocellular carcinoma in high-risk patients such as those with alcoholic ci...
Type : SEMANTIC
Result : UNKNOWN
Policy Ev. : Required: diagnosis: Evidence that the patient has hepatocellular carcinoma, such as a diagnosis code C22.0;
Required: underlying_condition: Evidence that the patient has one of the following high-risk conditions: alcoholic cirrhosis, cirrhosis of viral etiology, hemochromatosis, or alpha 1-antitrypsin deficiency
Patient Ev. : Hepatocellular carcinoma in high-risk patient with alcoholic cirrhosis.
Explanation : =====

AGENTIC SEMANTIC EVALUATION

Criterion: is covered for the diagnosis of hepatocellular carcinoma in high-risk patients such as those with alcoholic cirrhosis, cirrhosis of viral etiology, hemochromatosis, and alpha 1-antitrypsin deficiency
Policy: NCD NCD-190.25

Required Evidence:

- * diagnosis: Evidence that the patient has hepatocellular carcinoma, such as a diagnosis code C22.0
- * underlying_condition: Evidence that the patient has one of the following high-risk conditions: alcoholic cirrhosis, cirrhosis of viral etiology, hemochromatosis, or alpha 1-antitrypsin deficiency

Patient Evidence:

- * Hepatocellular carcinoma in high-risk patient with alcoholic cirrhosis.

Missing Evidence:

- * No evidence of cirrhosis of viral etiology, hemochromatosis, or alpha 1-antitrypsin deficiency
- * Required evidence not found in request: 'Diagnoses: C22.0'

Qwen Result: UNKNOWN
Critic Result: VALIDATED
Final Result: UNKNOWN

Insufficient or inconclusive evidence to determine whether this semantic criterion is satisfied. The result is UNKNOWN. Deterministic rules and EvidenceFusion will determine final coverage.

(Agentic pipeline completed in 60286ms)

Agent Trace:

[POLICY_AGENT] COMPLETED: Identified 2 required evidence categories in 29128ms.
[CLINICAL_EVIDENCE_AGENT] COMPLETED: Found 1 supporting, 0 contradicting, 2 missing evidence items in 18254ms.
Removed 1 unsupported inferences.
[EVALUATION_AGENT] COMPLETED: Partial evidence: 1 supporting items found, but 2 required evidence categories are missing.
Evidence is insufficient for a definitive conclusion.
[QWEN] COMPLETED: Qwen returned UNKNOWN. Evidence cited: 1 items.
[CRITIC_AGENT] COMPLETED: Critic VALIDATED Qwen's result: UNKNOWN. All 6 checks passed.

EVIDENCE FUSION

Criteria SATISFIED : 1
Criteria NOT_SATISFIED : 0
Criteria UNKNOWN : 1
Evidence Fusion Result : COVERED

FINAL DECISION

[APPROVE] Decision : APPROVE
Reason : All applicable policy criteria were satisfied. The submitted procedure meets the National Coverage Determination criteria. [NCD_CRITERIA_SATISFIED]
Reason Codes: NCD_CRITERIA_SATISFIED

Decision Basis:
All mandatory policy criteria were satisfied.
* NCD-NCD-190.25-HCPCS: SATISFIED
* NCD-NCD-190.25-C1474c5: UNKNOWN
Evidence Fusion: COVERED
DecisionEngine: COVERED -> APPROVE

APPROVE - TENS covered by NCD 160.7.1 [NCD RAG + NCD Decision Path]

REQUEST

Procedure Code : 64550
Diagnosis : G89.29
State : TX
Patient Age : 48
Clinical Notes : Patient presents with chronic pain syndrome following lumbar surgery. Conservative pharmacological therapy has

been tried for over 6 weeks without satisfactory relief. TENS (transcutaneous electrical nerve stimulation) requested as adjunct pain management.

POLICY IDENTIFICATION

NCD : N123
Jurisdiction : TX -> NOT_ADDRESSED

POLICY EVIDENCE

Source : NCD N123
Section : Coverage Indications
Text : Transcutaneous Electrical Nerve Stimulation (TENS) (HCPCS 64550) is covered for acute post-operative pain and chronic intractable pain when conservative therapy has been tried and failed. Documentatio
Similarity : 0.9000
Source : NCD N123
Section : Limitations
Text : TENS is not covered as initial therapy for acute pain; conservative measures must be attempted first. TENS for chronic non-cancer pain requires documented failure of conventional pharmacological treat
Similarity : 0.9000

CODE MATCHING

HCPCS 64550 -> NCD N123 -> MATCHED

POLICY CRITERIA

C1 [SQL] NCD-N123-HCPCS
Requirement : The requested procedure must be an applicable service under NCD N123.
Type : STRUCTURED
Result : SATISFIED
Policy Ev. : NCD N123 contains HCPCS 64550 in its covered-procedure list.
Patient Ev. : Submitted HCPCS: 64550
Explanation : Procedure 64550 is listed in NCD N123 covered HCPCS codes. Criterion SATISFIED by deterministic SQL check.

C2 [AGENTIC_QWEN] NCD-N123-C1766b5
Requirement : Documentation must support that the patient has a diagnosis of chronic pain requiring neurostimulation therapy...
Type : SEMANTIC
Result : UNKNOWN
Policy Ev. : Required: diagnosis_of_chronic_pain: Documentation of a diagnosis of chronic pain consistent with ICD-10 code G89.29 or equivalent; Required: inadequacy_or_contraindication_of_pharmacological_treatment: Documentation that conventional pharmacological treatment has been inadequate or contraindicated for the patient's chronic pain
Patient Ev. : Patient presents with chronic pain syndrome following lumbar surgery.; Conservative pharmacological therapy has been tried for over 6 weeks without satisfactory relief.; Transcutaneous Electrical Nerve Stimulation (TENS) (transcutaneous electrical nerve stimulation) requested as adjunct pain management.
Explanation : =====

AGENTIC SEMANTIC EVALUATION

Criterion: Documentation must support that the patient has a diagnosis of chronic pain requiring neurostimulation therapy and that conventional pharmacological treatment has been inadequate or contraindicated
Policy: NCD N123

Required Evidence:

- * diagnosis_of_chronic_pain: Documentation of a diagnosis of chronic pain consistent with ICD-10 code G89.29 or equivalent
- * inadequacy_or_contraindication_of_pharmacological_treatment: Documentation that conventional pharmacological treatment has been inadequate or contraindicated for the patient's chronic pain

Patient Evidence:

- * Patient presents with chronic pain syndrome following lumbar surgery.
- * Conservative pharmacological therapy has been tried for over 6 weeks without satisfactory relief.
- * Transcutaneous Electrical Nerve Stimulation (TENS) (transcutaneous electrical nerve stimulation) requested as adjunct pain management.

Missing Evidence:

- * Documentation of a diagnosis of chronic pain consistent with ICD-10 code G89.29 or equivalent is not explicitly stated in the patient data, though the diagnosis is listed in additional structured facts. The policy requires explicit documentation in the patient data, and this is not present.

Qwen Result: UNKNOWN
Critic Result: VALIDATED
Final Result: UNKNOWN

Insufficient or inconclusive evidence to determine whether this semantic criterion is satisfied. The result is UNKNOWN. Deterministic rules and EvidenceFusion will determine final coverage.

(Agentic pipeline completed in 65580ms)

Agent Trace:

- [POLICY_AGENT] COMPLETED: Identified 2 required evidence categories in 22211ms.
- [CLINICAL_EVIDENCE_AGENT] COMPLETED: Found 3 supporting, 0 contradicting, 1 missing evidence items in 27633ms.
- [EVALUATION_AGENT] COMPLETED: Partial evidence: 3 supporting items found, but 1 required evidence categories are missing. Evidence is insufficient for a definitive conclusion.
- [QWEN] COMPLETED: Qwen returned UNKNOWN. Evidence cited: 3 items.
- [CRITIC_AGENT] COMPLETED: Critic VALIDATED Qwen's result: UNKNOWN. All 6 checks passed.

C3 [AGENTIC_QWEN] NCD-N123-C166ffe
Requirement : patient has a diagnosis of chronic pain requiring neurostimulation therapy and that conventional pharmacologic...
Type : SEMANTIC
Result : UNKNOWN
Policy Ev. : Required: diagnosis: Evidence of a diagnosis of chronic pain, specifically coded as G89.29; Required: treatment_inadequacy_or_contraindication: Evidence that conventional pharmacological treatment has been inadequate or contraindicated for the patient's chronic pain
Explanation : =====
AGENTIC SEMANTIC EVALUATION

Criterion: patient has a diagnosis of chronic pain requiring neurostimulation therapy and that conventional pharmacological treatment has been inadequate or contraindicated
Policy: NCD N123

Required Evidence:

- * diagnosis: Evidence of a diagnosis of chronic pain, specifically coded as G89.29
- * treatment_inadequacy_or_contraindication: Evidence that conventional pharmacological treatment has been inadequate or contraindicated for the patient's chronic pain

Patient Evidence:

- * (No supporting evidence found in request)

Missing Evidence:

- * Evidence of a diagnosis of chronic pain, specifically coded as G89.29
- * Evidence that conventional pharmacological treatment has been inadequate or contraindicated for the patient's chronic pain

Qwen Result: UNKNOWN
Critic Result: VALIDATED
Final Result: UNKNOWN

Insufficient or inconclusive evidence to determine whether this semantic criterion is satisfied. The result is UNKNOWN. Deterministic rules and EvidenceFusion will determine final coverage.

(Agentic pipeline completed in 57703ms)

Agent Trace:

[POLICY_AGENT] COMPLETED: Identified 2 required evidence categories in 19619ms.
[CLINICAL_EVIDENCE_AGENT] FAILED: Clinical evidence extraction failed (timed out). All required evidence reported as missing.
[EVALUATION_AGENT] COMPLETED: No supporting evidence found. 2 required evidence categories are missing.
[QWEN] COMPLETED: Qwen returned UNKNOWN. Evidence cited: 0 items.
[CRITIC_AGENT] COMPLETED: Critic VALIDATED Qwen's result: UNKNOWN. All 6 checks passed.

C4 [AGENTIC_QWEN] NCD-N123-Cd9a571

Requirement : is covered for acute post-operative pain and chronic intractable pain when conservative therapy has been tried...

Type : SEMANTIC

Result : UNKNOWN

Policy Ev. : Required: pain_type: Patient must have either acute post-operative pain or chronic intractable pain; Required: conservative_therapy_tried: Documentation that conservative therapy was attempted

Explanation : =====

AGENTIC SEMANTIC EVALUATION

Criterion: is covered for acute post-operative pain and chronic intractable pain when conservative therapy has been tried and failed

Policy: NCD N123

Required Evidence:

- * pain_type: Patient must have either acute post-operative pain or chronic intractable pain
- * conservative_therapy_tried: Documentation that conservative therapy was attempted
- * conservative_therapy_failed: Documentation that conservative therapy failed to provide adequate relief

Patient Evidence:

- * (No supporting evidence found in request)

Missing Evidence:

- * Patient must have either acute post-operative pain or chronic intractable pain
- * Documentation that conservative therapy was attempted
- * Documentation that conservative therapy failed to provide adequate relief

Qwen Result: UNKNOWN
Critic Result: VALIDATED
Final Result: UNKNOWN

Insufficient or inconclusive evidence to determine whether this semantic criterion is satisfied. The result is UNKNOWN. Deterministic rules and EvidenceFusion will determine final coverage.

(Agentic pipeline completed in 53840ms)

Agent Trace:

[POLICY_AGENT] COMPLETED: Identified 3 required evidence categories in 17219ms.
[CLINICAL_EVIDENCE_AGENT] FAILED: Clinical evidence extraction failed (timed out). All required evidence reported as missing.
[EVALUATION_AGENT] COMPLETED: No supporting evidence found. 3 required evidence categories are missing.
[QWEN] COMPLETED: Qwen returned UNKNOWN. Evidence cited: 0 items.
[CRITIC_AGENT] COMPLETED: Critic VALIDATED Qwen's result: UNKNOWN. All 6 checks passed.

C5 [AGENTIC_QWEN] NCD-N123-Cadbeac

Requirement : must have failed conservative therapy for >= 6 weeks

Type : SEMANTIC

Result : UNKNOWN

Policy Ev. : Required: conservative_therapy_duration: Documentation showing the patient has undergone conservative therapy for at least 6 weeks and has failed to improve

Patient Ev. : Conservative pharmacological therapy has been tried for over 6 weeks without satisfactory relief.

Explanation : =====

AGENTIC SEMANTIC EVALUATION

Criterion: must have failed conservative therapy for >= 6 weeks

Policy: NCD N123

Required Evidence:

* conservative_therapy_duration: Documentation showing the patient has undergone conservative therapy for at least 6 weeks and has failed to improve

Patient Evidence:

* Conservative pharmacological therapy has been tried for over 6 weeks without satisfactory relief.

Missing Evidence:

* The patient data does not specify the type of conservative therapy that was administered or whether it included specific interventions (e.g., physical therapy, injections) beyond pharmacological therapy.

Qwen Result: UNKNOWN

Critic Result: VALIDATED

Final Result: UNKNOWN

Insufficient or inconclusive evidence to determine whether this semantic criterion is satisfied. The result is UNKNOWN. Deterministic rules and EvidenceFusion will determine final coverage.

(Agentic pipeline completed in 34643ms)

Agent Trace:

[POLICY_AGENT] COMPLETED: Identified 1 required evidence categories in 10588ms.
[CLINICAL_EVIDENCE_AGENT] COMPLETED: Found 1 supporting, 0 contradicting, 1 missing evidence items in 14887ms.
[EVALUATION_AGENT] COMPLETED: Partial evidence: 1 supporting items found, but 1 required evidence categories are missing.
Evidence is insufficient for a definitive conclusion.
[QWEN] COMPLETED: Qwen returned UNKNOWN. Evidence cited: 1 items.
[CRITIC_AGENT] COMPLETED: Critic VALIDATED Qwen's result: UNKNOWN. All 6 checks passed.

EVIDENCE FUSION

Criteria SATISFIED : 1

Criteria NOT_SATISFIED : 0
Criteria UNKNOWN : 4
Evidence Fusion Result : COVERED

FINAL DECISION

[APPROVE] Decision : APPROVE
Reason : All applicable policy criteria were satisfied. The submitted procedure meets the National Coverage Determination criteria. [NCD_CRITERIA_SATISFIED]
Reason Codes: NCD_CRITERIA_SATISFIED

Decision Basis:
All mandatory policy criteria were satisfied.
* NCD-N123-HCPCS: SATISFIED
* NCD-N123-C1766b5: UNKNOWN
* NCD-N123-C166ffe: UNKNOWN
* NCD-N123-Cd9a571: UNKNOWN
* NCD-N123-Cadbeac: UNKNOWN
Evidence Fusion: COVERED
DecisionEngine: COVERED -> APPROVE

REQUEST_MORE_INFORMATION - Outside jurisdiction (CA) [LCD Path]

REQUEST

Procedure Code : 64483
Diagnosis : M54.17
State : CA
Patient Age : 63
Clinical Notes : Lumbosacral radiculopathy.

POLICY IDENTIFICATION

NCD : NOT_ADDRESSED
Jurisdiction : CA -> NOT_MATCHED

POLICY EVIDENCE

RAG : NOT USED - no semantic criterion identified

EVIDENCE FUSION

Criteria SATISFIED : 0
Criteria NOT_SATISFIED : 0
Criteria UNKNOWN : 0
Evidence Fusion Result : NOT_ADDRESSED

FINAL DECISION

[INFO] Decision : REQUEST_MORE_INFORMATION
Reason : The request is missing required information: State outside jurisdiction. Additional information must be provided before a coverage determination can be made. [MISSING_REQUIRED_INFORMATION]

Reason Codes: MISSING_REQUIRED_INFORMATION

Decision Basis:

Additional information is required to determine coverage.

* No formal criteria were evaluated (deterministic code matching applied).

Evidence Fusion: NOT_ADDRESSED

DecisionEngine: NOT_ADDRESSED -> REQUEST_MORE_INFORMATION

REQUEST_MORE_INFORMATION - No matching policy

REQUEST

Procedure Code : 99999

Diagnosis : M54.16

State : TX

Patient Age : 45

Clinical Notes : Unknown procedure code.

POLICY IDENTIFICATION

NCD : NOT_ADDRESSED

Jurisdiction : TX -> NOT_ADDRESSED

POLICY EVIDENCE

RAG : NOT USED - no semantic criterion identified

EVIDENCE FUSION

Criteria SATISFIED : 0

Criteria NOT_SATISFIED : 0

Criteria UNKNOWN : 0

Evidence Fusion Result : NOT_ADDRESSED

FINAL DECISION

[PEND] Decision : PEND

Reason : Policy evidence is currently unavailable (CMS API error). Manual review required.

Reason Codes: EVIDENCE_UNAVAILABLE

Decision Basis:

Policy evidence is currently unavailable (CMS API error). Manual review required. Evidence Fusion: NOT_ADDRESSED.

DecisionEngine: NOT_ADDRESSED -> PEND.

END OF DEMO
